# Adding theophylline to ICS therapy on COPD: a protocol for systematic review and meta-analysis

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## Citation

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## Review question

To study the efficacy and safety of adding theophylline to ICS therapy in the treatment of COPD patients

## Searches

PubMed, Web of Science, Cochrane Library, Embase

## Types of study to be included

Any comparative study, i.e., cohort, case-control, RCTs

## Condition or domain being studied

Chronic obstructive pulmonary disease (COPD) is a common and frequently occurring respiratory disease. A preventable and treatable disease characterized by limited airflow, associated with abnormal inflammatory responses of the lungs to harmful particles or gases, and COPD may also affect changes in various systems throughout the body, thus having systemic effects.

## Participants/population

Patients with chronic obstructive pulmonary disease

## Intervention(s), exposure(s)

theophylline to ICS therapy

## Comparator(s)/control

ICS therapy

## Main outcome(s)

Severe COPD exacerbations, COPD hospital admissions, health-related quality-of-life score, total admissions, FEV1, FVC, MVV, CAT, mMRC, pneumonia, all-cause mortality, COPD-related mortality and adverse effects.

## * Measures of effect

Mean difference, odds ratio (OR) or relative risks (RR)

## Additional outcome(s)

None

## * Measures of effect

None

## Data extraction (selection and coding)
Author, publication year, country, study design, characteristics of participants(i.e., age[MD±SD], sex), treatment period, dosage, sample size(n, cases, control), effect size(WMD, SMD, RR, OR, HR with their 95%CI), outcome indicator value(i.e., exacerbations, FEV1, FVC, MVV, CAT, mMRC, pneumonia, mortality, adverse effects, mortality).

## Risk of bias (quality) assessment
Cochrane collaboration's tool forassessing risk of bias, NOS scale(cohort or case-control study), AHQR scale(cross-sectional study)

## Strategy for data synthesis
We calculated the results of individual studies as fixed eHect weighted mean diference (WMD) or standardised mean diference (SMD) including the 95% confidence interval (CI) for each outcome. Where results were expressed as dichotomous variables, we calculated odds ratio (OR) or relative risks (RR) with 95% CI for individual outcomes. Stata/SE 15.0 software was used for statistical analyses. Heterogeneity between studies was assessed using the Higgins $I^2$ test and P values, with $P < 0.05$ indicating the significant heterogeneity between studies, and the $I^2$ statistic which describes the percentage of total variation across trials that is due to heterogeneity rather than sampling error. The values of 25%, 50%, and 75% for the $I^2$ as indicative of low, moderate, and high statistical heterogeneity, respectively. If $P < 0.05$ and $I^2 > 50\%$, the random effects model was selected to calculate the pooled effective size. Publication bias was assessed with funnel plots and sensitivity analyses were performed for studies that have a greater impact on the results. All tests were two tailed and $P < 0.05$ was considered significant.

## Analysis of subgroups or subsets
When heterogeneity is high and data is supported, we will consider doing subgroup analysis, such as, country, study design, dosage.

## Contact details for further information
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Lanzhou university

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## Type and method of review
Intervention, Meta-analysis, Systematic review

## Anticipated or actual start date
20 December 2020

## Anticipated completion date
01 June 2021

## Funding sources/sponsors
No

## Conflicts of interest

## Language
English

## Country

China

## Stage of review
Review Completed published

## Subject index terms status
Subject indexing assigned by CRD

## Subject index terms
MeSH headings have not been applied to this record

## Date of registration in PROSPERO
13 January 2021

## Date of first submission
13 December 2020

## Stage of review at time of this submission

| Stage | Started | Completed |
|---|---|---|
| Preliminary searches | Yes | No |
| Piloting of the study selection process | Yes | No |
| Formal screening of search results against eligibility criteria | Yes | No |
| Data extraction | No | No |
| Risk of bias (quality) assessment | No | No |
| Data analysis | No | No |

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*The record owner confirms that they will update the status of the review when it is completed and will add publication details in due course.*

## Versions
13 January 2021
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